Response to Reviewers

Manuscript: PCLM-D-26-00445

Title: Kaya-guided hybrid deep learning for leave-country-out estimation of national CO_2 emissions: a 112-country benchmark with country-clustered inference, conformal uncertainty, and explainable AI

Journal: PLOS Climate

Corresponding author: Samiuel C. Atuahene

Dear Dr Prina,

We thank you and the two reviewers for the careful reading of our manuscript. The comments have led to a substantially restructured and, we believe, much clearer paper. We respond to every point below. Reviewer comments are quoted in italics, our responses follow in plain text, and each response states where the change can be found in the revised manuscript. Because the manuscript was reorganized, we refer to sections, tables and figures of the revised version by name and number; line numbers in the reviewers' comments refer to the original submission.

Summary of the main changes

1. **Structure.** The Introduction and the former Related work section were merged and shortened, and the aims are now stated as four research questions (Q1–Q4). A new section, *Study design and workflow*, with a flowchart (Fig 1) introduces the three experiments before they are used. Methods, Results and Discussion are now strictly separated: the Results contain no methods and no interpretation.
2. **Density.** The main text now contains 7 numbered equations instead of 34 and 8 figures instead of 12 (plus 3 that were marked as supplementary). The supporting temporal experiment (Experiment B), the equations of the standard network layers, the algorithms

and the notation table were moved to Supporting Information (S1 Text, S2 Text). The appendix tables were replaced by two supplementary workbooks (S1 Table, S2 Table).

3. **Main message.** The central finding, that neither deep nor hybrid models were shown to outperform ridge regression, is now stated plainly in the Abstract, Results, Discussion and Conclusions. The Discussion has a new structured section on why deep learning did not outperform the linear model and a roadmap of recommendations.
4. **New analyses.** We added (i) sensitivity analyses of the missing-data treatment, (ii) analyses without the seven atypical economies, (iii) two further measures of covariate importance defined on held-out error (permutation and drop-one-covariate), (iv) a like-for-like test of the value of the low-carbon share with identical countries and folds, (v) an analysis of conformal coverage for ridge regression that explains the coverage shortfall, and (vi) country-clustered tests of ridge regression against each classical learner.
5. **Data description.** A table of variables with units, sources, summary statistics and the amount of filling (Table 1) and a figure with the time series of the target and all covariates (Fig 2) were added.

Corrections found while re-running the analyses

After the first revision we re-ran the whole pipeline from the archived code, in a clean copy, and re-checked the numbers against the manuscript. The following corrections resulted. None of them changes the conclusions, although two statements in the first draft were too strong and are now corrected.

9. *Size of the inner validation set.* The text said that one-seventh of the training countries (13) formed the inner validation set. The code takes one-seventh rounded down, that is 12 countries in Experiment A and 8 in Experiment C. This is now stated in the Methods and

S1 Text, and the calibration analysis for ridge regression was re-run with 12 countries (average coverage 87.4% instead of 87.2%; single folds range from 45% to 100%).

10. *Prediction intervals.* The intervals reported in the first draft belong to the hybrid of Experiment C (E-PSO-KG-CBA, 76 countries); the Methods, Results and the caption of the South Africa and Egypt figure now say so. The script that produced them (`conformal.py`) was missing from the archive. It has been rebuilt from the method in the manuscript; the rebuilt fold networks reproduce the archived predictions exactly, and across repeated Monte-Carlo runs the coverage lies between 81.0% and 83.0% (archived: 81.9%).

11. *Like-for-like coverage for ridge regression.* With the 8 calibration countries used for the hybrid, ridge regression in Experiment C reaches 86.5% coverage on average; calibration on all training countries by grouped cross-conformal prediction gives 88.8%, and an exchangeable reference gives 90.3% (Results, *Prediction intervals*; S2 Table). The coverage of the hybrid's intervals is therefore not better than that of a simple linear model calibrated in the same way.

12. *Significance of the deep networks.* The first draft said that unguided deep networks were significantly less accurate than ridge regression. Only the two convolutional networks are (CNN-LSTM, $p = 0.001$; CNN-BiLSTM-Attn, $p = 0.002$); LSTM ($p = 0.08$) and GRU ($p = 0.12$) are not. Table 4 now gives the cluster-bootstrap tests of KG-CBA (+0.48 points, $p = 0.50$) and PSO-KG-CBA (+0.47, $p = 0.51$) against ridge regression, which the first draft had left as a pending item.

13. *Error without the atypical economies.* The first draft derived the RMSE of the hybrids without the seven atypical economies from error shares. We now report the values

computed directly from the stored predictions: RMSE 1.48 for ridge regression, 1.75 for KG-CBA, 1.80 for PSO-KG-CBA and 1.45 for the ensemble. The seven economies account for 84.3% (ensemble) and 86.5% (PSO-KG-CBA) of the squared error.

14. *South Africa and Egypt.* The text gave per-country MdAPE values that did not match the panel titles of the figure, because they refer to different models. The text now states the model: for the Experiment C ensemble 6.1% (South Africa) and 3.4% (Egypt); for the hybrid 5.6% and 5.7%; for ridge regression 5.5% and 4.9%.
15. *Settings.* Table 2 now gives the LightGBM settings of Experiment A and the additional settings of Experiment C, and the untuned KG-CBA settings, which were previously marked as to be confirmed.
16. *Reproducibility.* The archived particle swarm log was missing; the search was re-run and reproduces the published configuration as its best solution, with fitness values that match the original figure. The classical models and the Experiment C hybrid reproduce from a clean copy; the Experiment A deep models vary between runs (MdAPE up to 0.3 points), and the country-clustered conclusions are unchanged (the comparison is given in the README and REPRODUCE.md of the repository).

Corrections made on our own initiative

While preparing the revision we re-derived the panel from the public source files and re-checked every statement. We found the following errors and imprecisions in the original submission and have corrected them. None of them changes the conclusions.

1. *Correlations.* The text gave correlations of 0.93 and 0.82 between the logarithm of per-capita emissions and the logarithms of energy and GDP per capita. The correct values,

which the original figure already showed, are 0.97 and 0.88 (Results, *Data overview*; S2 Fig).

2. *Description of the fill rule.* The original text said that only gaps of up to three years were interpolated and that series edges were extended. In fact, longer gaps inside a series were also completed (three years by interpolation, the remainder by carrying the last value forward). The rule is now described exactly (Methods, *Data*).
3. *Nature of the long filled runs.* We had interpreted the long filled runs in the coal and gas shares as evidence of thin statistics, and we had stated that heavily filled countries are easier to estimate, so that the full-panel results were mildly optimistic. Neither statement holds. The blank entries are zeros for fuels that a country did not burn (in all 464 affected country-years the reported national total equals the sum of the reported components to within 0.3%), and countries with long filled runs are neither consistently easier nor consistently harder to estimate (Table 5). The corresponding passages in the Results, Discussion and Conclusions were rewritten.
4. *Like-for-like comparison in Experiment C.* The original six-covariate values for the 76 countries (median error 14.3% for ridge regression) came from the models of Experiment A, trained on 112 countries with different folds. We now also report the comparison with identical countries and folds (15.6% against 7.7%), with a country-clustered test (Table 4, Table 6).
5. *Data sources.* The data files used (accessed on 26 July 2026) redistribute the Global Carbon Budget 2025, the Statistical Review of World Energy 2025, the International Energy Data of the U.S. Energy Information Administration and the Maddison Project Database 2023. The original manuscript cited the 2024 editions and did not cite the last

two sources. The citations were corrected and the exact file versions are given (Methods, *Data*).

6. *List of countries with long filled runs.* The former Table B5 omitted several countries. S1 Table now gives the complete audit for all 112 countries.
7. *Kuwait 1991.* The original text said that the extreme value for Kuwait in 1991 enters “as covariate context”. Per-capita emissions are never a covariate; the sentence was corrected.
8. *Reference list.* Every reference was checked against the publisher’s record. Incomplete entries were completed, and references whose bibliographic details we could not verify were removed. References supporting the statements queried by Reviewer 2 were added. All authors have confirmed.

Response to the journal requirements

1. AI tool use statement.

The statement was rewritten to give the name of the tool, how it was used, how the validity of its outputs was evaluated and which parts of the work were affected. It is placed in the Acknowledgments and reads:

> Use of artificial intelligence tools. The authors used the generative AI tool Claude (Anthropic, San Francisco, CA, USA; Claude Fable 5.2 (API: claude-fable-5-2)) in three ways. (i) Code: Claude drafted and revised Python code for data harmonization, model training, statistical tests and figures, following analysis specifications written by the authors. (ii) Analyses added in revision: the scripts for the sensitivity, importance and conformal analyses were drafted and first run with Claude from the public source files. (iii) Text: Claude assisted with drafting and language editing of the manuscript

and of the response to the reviewers. No AI tool generated or altered data; all data come from the cited public sources. To evaluate the validity of the outputs, the authors read and executed all code and regenerated every table and figure from the released scripts; confirmed that independently re-implemented deterministic models reproduce the reported values to three decimal places; checked every numerical statement against the script output and every reference against the publisher's record; and reviewed and edited all text. The authors take full responsibility for the content of this article.

2. Code sharing.

All author-generated code is available without restriction in a public repository that is archived with a DOI: insert DOI of the archived release. It contains the code that builds the panel from the public source files, all models, the out-of-fold estimates of every model, the scripts that produce each table and figure, and the scripts of the analyses added in revision. Software and package versions are given in the Methods (*Software, data and code*).

3. Competing interests statement.

All authors have confirmed. I have read the journal's policy and the authors of this manuscript have the following competing interests: S.C.A. is a director of Premium Intelligence Networks and Systems Co. Ltd., Accra, Ghana, a private company. The company had no role in the study design, data collection and analysis, decision to publish, or preparation of the manuscript, and has no commercial interest in the results. This does not alter our adherence to PLOS Climate policies on sharing data and materials. The other authors have declared that no competing interests exist.

4. Financial disclosure statement.

All authors have confirmed. The authors received no specific funding for this work.

5. Figure files.

All eight figures are supplied as separate TIFF files (Fig1.tif to Fig8.tif; RGB, 300 dpi, LZW compression, at most 2250 pixels wide), and the figures are no longer embedded in the manuscript file. All authors have confirmed.

6. Legends of Supporting Information files.

A full list of legends for the Supporting Information files (S1 Text, S2 Text, S1 Table, S2 Table, S1–S6 Fig) was added after the reference list.

7. Data repository.

The minimal data set that is needed to replicate the findings (the harmonized panel, the out-of-fold estimates of every model and the values behind every table and figure) has been deposited in a public repository with a DOI. Our revised Data Availability statement is:

> All data underlying the findings are publicly available. The harmonized panel, the out-of-fold estimates of every model, the values behind every table and figure and all code are available from https://github.com/samfozzy/kg-cba-pclm-revision, archived with the DOI insert DOI. The source data are the Our World in Data CO_2 and energy data sets (https://github.com/owid/co2-data and https://github.com/owid/energy-data; repository versions of 2 June 2026 and 27 April 2026), which redistribute the Global Carbon Budget 2025, the Energy Institute Statistical Review of World Energy 2025, the International Energy Data of the U.S. Energy Information Administration and the Maddison Project Database 2023.

We have also asked all co-authors to verify their ORCID iDs in the submission system.

Response to Reviewer #1

I have read the article […]. The study considers a large number of models and analyses, which makes the paper difficult to follow. There is also a noticeable gap between the study objectives, methodology, and presented results. In particular, the statistical analysis of the input data is limited, and the methodological steps and model-selection process are not clearly demonstrated through a structured workflow or flowchart. Therefore, the overall methodology and presentation require substantial improvement.

We thank the reviewer for this assessment, which we accept. The revised manuscript links objectives, methods and results explicitly: the Introduction ends with four research questions, the workflow figure (Fig 1) shows which experiment and which analysis answers each question, and the Discussion opens by answering the four questions in turn. The input data are now described statistically (Table 1, Fig 2, S1 Table, S2 Fig). The individual points are addressed below.

Comment 1. *The abstract contains too much information, making it difficult to follow. Please rewrite and simplify the abstract so that readers can clearly understand the paper. Focus mainly on the study goal, methodology, key results, and practical implications.*

Response. The Abstract was rewritten in that order: the goal (two sentences), the method (two sentences), the key results (five sentences) and the practical implication (one sentence). Secondary results (conformal coverage, the temporal experiment, the comparison of testing regimes) and most of the statistics were removed from it.

Comment 2. *One of the major concerns is the readability of the methodology section. […] The authors should provide a clear flowchart illustrating the research problem, models used, validation procedures, model comparison and selection, and the implications of the final results.*

Response. We added a flowchart (Fig 1) with seven steps: research questions, data, experiments, models, validation and comparison (including the selection rule), diagnostics and robustness, and outcome and implications. The Methods section follows the same order: *Study design and workflow*, *Data*, *Estimation task and validation*, *Models*, *Model comparison and selection*, *Sensitivity analyses*, *Prediction intervals*, *Explainability*, and *Software, data and code*. Standard material (the equations of the network layers, the algorithms and the notation table) was moved to S1 Text, so that the Methods contain only what is specific to this study.

Comment 3. *In Table 1, the proposed deep-learning models do not clearly outperform the simpler Ridge model. In fact, Ridge achieves better RMSE and R^2 values than other models. The authors should clearly explain the practical advantages of using these more complex models and justify why they were selected or proposed.*

Response. The reviewer is right, and this is the main finding of the study. The original text did not state it plainly enough; the revision does so in the Abstract, the Results, the Discussion and the Conclusions. Specifically:

- *Practical advantages.* We found none in accuracy and say so (Discussion, *Why deep learning did not outperform the linear model*, penultimate paragraph). We recommend ridge regression as the estimator for inventory gap-filling.
- *Why the complex models were included.* They represent current practice in the literature and had to be tested under a valid protocol before they could be set aside. The Kaya-guided hybrid was proposed to test one specific question: whether supplying the linear structure repairs the deficit of deep networks. It does (median error 20.8% without guidance against 15.7% with it, $p = 0.002$), which shows that the deficit is a failure to learn the linear structure from few countries and not a lack of capacity; but nothing

remains for the network to add. This reasoning is now given in the Introduction and the Discussion.

- *RMSE and R^2.* We explain in the Methods (*Model comparison and selection*) why the median absolute percentage error is the primary measure: per-capita emissions are strongly right-skewed, so RMSE and R^2 are dominated by a few high emitters. A new analysis shows this directly: the low R^2 of the hybrids (0.52–0.55) is produced by a handful of countries, but removing those countries still does not make the hybrids more accurate than ridge regression (Results, *Error concentration in atypical energy systems*; Table 5; Fig 5).
- *Selection.* Ridge regression is retained as the reference estimator under an explicit rule (see Comment 7).

Comment 4. *What is the size of the dataset used in this study? The authors should clearly report the number of observations, countries, and years included in the analysis. In addition, the time series of all input (independent) variables and the output (dependent) variable should be presented […].*

Response. The size of the data set is now stated in the Methods (*Data* and *Estimation task and validation*): 112 countries observed for 34 years (1990–2023), that is 3,808 country-years, which give 27 modelling samples per country (target years 1997–2023) and 3,024 samples in total; the enriched panel of Experiment C has 76 countries and 2,052 samples. A new table (Table 1) lists every variable with its unit, source, median and interquartile range and the amount of filling. A new figure (Fig 2) shows the time series of the target and of every covariate as the median and the 25th–75th and 10th–90th percentiles across countries. S1 Table lists the 112 countries, and S2 Fig gives the correlations between the variables.

Comment 5. *In Table 1, the first model is listed as "Mean." Is "Mean" intended to represent a model or simply a baseline mean predictor? Please clarify this in the table and methodology.*

Response. It is a naive baseline. It is now called "Mean predictor", its family is given as "Naive baseline" in Tables 2 and 3, and the Methods (*Benchmark models*) state that it "is not a fitted model in the usual sense: it assigns to every test sample the mean of y over the training countries and shows the error obtained without any covariate information".

Comment 6. *Which software and programming environment were used to implement and evaluate all the models? Please clearly specify the software, packages, and versions used. The authors should also provide the source codes for all implemented models to ensure reproducibility of the study.*

Response. A section *Software, data and code* was added to the Methods. It gives the Python version and the versions of TensorFlow, scikit-learn, XGBoost, LightGBM and SHAP for the original run and for the analyses added in revision. The source code of all models, the code that builds the panel, the out-of-fold estimates and the scripts for every table and figure are available in the public repository https://github.com/samfozzy/kg-cba-pclm-revision, which is archived with the DOI insert DOI.

Comment 7. *In Tables 1, 4, and 6, several models show good performance, and the large number of models makes the results difficult to follow. Since the methodology and model selection process are not clearly demonstrated, there appears to be a gap between the study objectives, methodology, and presented results. The authors should clearly explain why these models were selected, how they were evaluated and compared, and how the final or best-performing models were identified.*

Response. Three changes address this point.

- *Why these models.* The Methods (*Benchmark models*) now explain that the thirteen models span the families used in the emission-modelling literature in increasing order of flexibility: a naive baseline, one linear model, five classical machine-learning models, four deep sequence networks that are the building blocks of published hybrids, and the two Kaya-guided hybrids. Table 2 gives the family and the settings of each model, and Table 3 and Fig 4 group the models by family.
- *How they were compared and selected.* The new section *Model comparison and selection* states the primary accuracy measure, the three statistical tests and the selection rule: "The reference estimator was taken to be the simplest covariate-based model that no other model was shown to outperform significantly in the cluster bootstrap." The Results apply this rule in the order accuracy (Table 3), tests (Table 4), decision.
- *Fewer models in view.* The former Table 6 (temporal forecasting) was moved, with Experiment B, to S2 Text. The ensemble is marked as exploratory in every table and is not a candidate for selection.

Comment 8. *Some input variables, particularly coal, oil, and gas shares, are derived from CO_2 data. This raises questions about using the model in countries where CO_2 inventories are unavailable. The ablation analysis helps, but this issue needs clearer discussion.*

Response. We now discuss this in four places. The Methods (*Estimation task and validation*, last paragraph) state what the fuel shares are, that they carry no information on the level of emissions, and that they would have to be derived from energy balances for a country without emission accounts. The Results (*Robustness to missing-data treatment and covariate set*; Table 5) report the model without them, now with a test: the median error is 16.9% against 16.3%, a difference that is not significant ($p = 0.22$). The Discussion (*What limits accuracy: covariate*

data) concludes that the estimator "can therefore be applied to a country that has energy and economic statistics but no emission inventory", and the roadmap recommends deriving fuel shares from energy balances (item 3). The Limitations add that the remaining covariates also come from the statistical systems that produce inventories, so that the estimator is a cross-check and a means of filling gaps, not a substitute for national energy statistics.

Comment 9. *The imputation approach is concerning. Some variables were extended for very long periods, reaching up to 31 years. A sensitivity analysis excluding countries with extensive imputation would strengthen the results.*

Response. We share the concern and examined it in depth. This led to a correction of our own interpretation (see *Corrections made on our own initiative*, item 3).

- *What the long runs are.* They occur only in the coal and gas shares, and they are not unknown values. In the source, the emissions of a fuel are blank where that fuel does not appear in the national total: in all 464 country-years with a blank coal or gas entry, the reported total equals the sum of the reported components to within 0.3%. The blanks are therefore zeros for fuels that a country did not burn (for example, the gas entries for El Salvador are blank until 2020). The fill rule had replaced these zeros with small positive values: 81% of the filled coal shares and 68% of the filled gas shares are below 0.01 (Methods, *Data*).
- *Sensitivity analysis excluding countries with extensive imputation.* As requested, we repeated the analysis without the 25 countries that have a filled run of more than three years and without the 19 countries that have a run of more than ten years. The median error of ridge regression is 17.6% and 15.6%, against 16.3% for the full panel (Table 5).

- *Further analyses.* Setting the blank shares to zero instead of filling them changes the median error of ridge regression from 16.3% to 16.2% ($p = 0.19$) and that of each classical learner by at most 0.3 percentage points. Scored separately, the 87 countries without long filled runs have a median error of 15.5% and the 25 countries with long runs 18.3% (Table 5; S2 Table). The model without fuel shares, which needs no long fill at all, has a median error of 16.9%.

The results are therefore robust to the treatment of the missing values. The neural networks were not re-trained under the alternative treatments; this is stated in the Methods and the Limitations.

Comment 10. *The uncertainty analysis is not clear.*

Response. The description was rewritten (Methods, *Prediction intervals*). It now explains the construction in three numbered steps in plain language (Monte-Carlo dropout, split-conformal calibration, back-transformation), defines the two measures by which the intervals are judged (coverage and mean width), and gives the formulas with all symbols in S1 Text (section F). The Results (*Prediction intervals*) first report the coverage before and after calibration and then explain the remaining shortfall with a new analysis for ridge regression: coverage is 90.2% when calibration and test samples come from the same countries, as the theory requires, and falls below 90% and varies strongly between folds when they come from different countries. The interpretation was moved to the Discussion (Limitations; roadmap item 6).

Comment 11. *Consider adding the hyperparameters of all considered models.*

Response. Table 2 now lists the settings of all thirteen models and of the ensemble, the common training settings of the deep networks and the search ranges and selected values of the particle swarm optimization.

Comment 12. *The PSO optimization uses only 30 evaluations, which seems limited for tuning a relatively complex deep-learning model. The authors should better justify this choice or examine the sensitivity of the results to the optimization procedure.*

Response. We agree that the budget is small, and we now justify the choice and show that the conclusions do not depend on the optimization.

- *Justification.* Each evaluation trains a network, and the fitness is computed on 12 validation countries in Experiment A and 8 in Experiment C (Methods, *Kaya-guided hybrid network*).
- *Sensitivity of the results to the optimization.* The hybrid with fixed, untuned settings is reported next to the tuned one in every table. Tuning did not improve it: the median error is 15.7% for the untuned and 16.7% for the tuned hybrid, and both are significantly more accurate than the unguided network ($p = 0.002$ for each; Table 4). The conclusions on Kaya guidance and on ridge regression are therefore the same with and without the optimization.
- *Why a larger budget would not help.* The Discussion (*Why deep learning did not outperform the linear model*, "Tuning added variance, not accuracy") explains that a fitness value computed on 12 countries is noisy, so that selecting the best of many configurations mostly selects noise; this limitation comes from the number of countries and not from the size of the search.
- *Limitations.* The Limitations now state the small budget and one further point that we identified ourselves: the configuration selected on the first fold was used in the other folds, whose test countries had been part of the search data. As tuning did not improve accuracy, this cannot have inflated the results.

Particle swarm optimization is no longer presented as a contribution of the paper.

Comment 13. *The ensemble was selected after examining the out-of-fold results. Therefore, its reported performance may be optimistic and should be presented as exploratory rather than a fully independent evaluation.*

Response. We agree and have followed the suggestion throughout. The ensemble is now labelled "exploratory" in the Methods ("an exploratory analysis whose accuracy is likely to be optimistic, and it was not a candidate for the reference estimator"), in Tables 2, 3 and 6, in Figs 1 and 4, in the Results and in the Limitations.

Comment 14. *The manuscript is very dense, with many models, experiments, equations, statistical tests, and additional analyses. The paper would benefit from substantial shortening and restructuring to make the main objectives, methodology, and findings easier to follow.*

Response. The manuscript was restructured as described under *Summary of the main changes*. In numbers: the main text has 7 numbered equations instead of 34; 8 figures instead of 12, with 6 further figures in Supporting Information; no separate Related work section; and one experiment fewer, because Experiment B is now in S2 Text. The notation table, the algorithms and the extended tables are in Supporting Information. We should say openly that the running text is not shorter in words (about 7,100 words against about 6,200), because both reviewers also asked for additional analyses, for definitions of all symbols and abbreviations, and for a fuller discussion. We have tried to make the additional text easy to follow by organizing it around the four research questions.

Response to Reviewer #2

We thank the reviewer for the detailed annotations, which were very helpful. We first respond to the two major comments and then to each of the 36 comments in the annotated file, in the order in which they appear.

Major comments

Major comment 1. *The most important finding of this study is that using (hybrid) deep learning models do not significantly outperform a simple ridge regression in estimating CO2 emissions. Throughout the paper several causes are mentioned, but it could improve the paper if it is addressed in a more clear and structured way in the discussion. Additionally, a roadmap with specific recommendations on what is needed to make better estimates in future could be included.*

Response. We agree on both points.

- *The finding.* It is now stated plainly in the Abstract ("no network or ensemble was shown to be more accurate than ridge regression"), in the Results (*Country-clustered comparison of models*, which ends with the retention of ridge regression as the reference estimator), at the start of the Discussion (answer to Q2) and in the Conclusions. The cluster-bootstrap tests of KG-CBA and PSO-KG-CBA against ridge regression are now reported in the text and in Table 4 (+0.48 points, $p = 0.50$, and +0.47, $p = 0.51$).
- *Causes.* The Discussion has a new section, *Why deep learning did not outperform the linear model*, which gives five causes, each tied to a result of the study: (1) the relationship is close to linear by construction (Kaya identity; a model with energy per capita alone reaches a median error of 17.9%); (2) the number of independent units is the number of countries, about 90 per training set, and not the number of country-years; (3)

held-out countries can lie outside the range of the training countries, where tree ensembles cannot extrapolate and networks behave unpredictably; (4) the remaining error is caused by missing information and not by the functional form, as Experiment C shows; and (5) tuning adds variance and not accuracy when the fitness is computed on 12 countries.

- *Roadmap.* The Discussion has a new section, *A roadmap for better national estimates*, with seven specific recommendations in order of expected benefit: extend energy-mix statistics; add covariates for the atypical energy systems; derive fuel shares from energy balances; keep the linear core and test additions against it; adopt evaluation standards that match the task; calibrate uncertainty on many countries; and extend the scope to sectors, sub-national units and other gases.

Major comment 2. *The paper structure needs improvement. Some parts in the introduction are highly repetitive, and the results sections contains elements that would fit better in either the methods or in the discussion. Moreover, some sentences use informal language. Acronyms/abbreviations and variables should be written out and/or explained at first mention.*

Response. All four points were addressed.

- *Repetition.* The Introduction and the Related work section were merged (about 700 words instead of about 1,260), and the list of contributions and the section “The gap this study fills”, which repeated each other, were replaced by four research questions.
- *Methods and interpretation in the Results.* The description of the experiments, the workflow and architecture figures, the imputation audit and the rationale of the tests are now in the Methods; interpretive statements were moved to the Discussion. The Results report findings only.

- *Informal language.* The text was rewritten in a neutral register throughout; expressions such as “the honest one”, “decisively” and “sharpens the same point from an unexpected direction” were removed.
- *Abbreviations and variables.* Every abbreviation is written out at first use in the text and in the legend of every figure and the footnote of every table in which it appears. Every symbol is defined where it first occurs, and S1 Text contains a complete notation table.

Comments in the annotated manuscript

Line numbers refer to the original submission.

1. Lines 35–36 (Abstract). *“In conventional temporal forecasting, learners beat naive persistence only marginally and inconsistently across evaluation windows.” – Which learners is referred to? Sentence seems a bit out of place.*

Response. We agree that the sentence was out of place. The temporal experiment is a supporting analysis and has been moved to S2 Text; it is no longer mentioned in the Abstract. S2 Text names the learners (ridge regression, random forest, XGBoost and LightGBM) and the two naive baselines.

2. Line 28 (Abstract). *“where observation-level testing would report $p < 10^{-9}$” – What do you mean?*

Response. We meant a test that treats each of the 3,024 country-years as an independent observation, in contrast to a test that treats the 112 countries as the independent units. The phrase was removed from the Abstract. The two kinds of test are now defined in the Methods (*Model comparison and selection*), where the first is introduced as “the observation-level test”, and the contrast is reported in the Results (*Country-clustered comparison of models*, second paragraph).

3. Lines 58–63 (Introduction). *"First, most studies evaluate temporal forecasting […]. Second, validation protocols frequently allow the same country to appear in both training and test data […]." – Reference?*

Response. References were added for each statement: examples of studies that forecast for countries with observed histories (Ajala et al.; AlShehri; Kong et al.; Tian et al.), and the methodological literature on validation of grouped data and on leakage (Roberts et al. 2017; Kapoor and Narayanan 2023). The third statement, on naive benchmarks, now cites Hyndman and Koehler (2006) and Makridakis et al. (2018).

4. Line 65 (Introduction). *"Fourth, […] significance is almost always assessed by treating every country-year as an independent observation." – Reference?*

Response. We now cite the standard references on inference with clustered data (Moulton 1990; Bertrand, Duflo and Mullainathan 2004; Cameron and Miller 2015). The wording was also moderated from "almost always" to "usually".

5. Lines 70–71. *LOCO is also used for Leave-one-covariate-out analyses. A way of assessing feature importance.*

Response. Thank you for pointing this out. To avoid the ambiguity we now abbreviate leave-country-out as LCO and say why (Introduction, fourth paragraph). This matters all the more because the revision adds a leave-one-covariate-out analysis (drop-one-covariate importance), for which we cite Lei et al. (2018).

6. Line 73. *"persistence is undefined" – You mean persistence is not present?*

Response. Yes. The sentence now reads: "Because the target country contributes nothing to model fitting, no persistence forecast can be formed for it".

7. Lines 86–87. *"The contributions are fivefold, and are as much about how the evidence is established as about what it shows." – This sentence is a bit unclear.*

Response. The sentence was removed. The aims are now stated as four research questions at the end of the Introduction.

8. Lines 91–92. *"a handful of structurally atypical energy systems dominates" – I.e. a couple of countries. Have you tried running the model and calculating performance with these countries excluded?*

Response. We have now done so (Methods, *Sensitivity analyses*; Results, *Error concentration in atypical energy systems*; Table 5; Fig 5; S2 Table). The seven economies are named. For ridge regression, leaving them out of the scoring lowers the RMSE from 3.06 to 1.48 t per person and raises R^2 from 0.80 to 0.90, while the median error changes only from 16.3% to 15.1%; re-estimating the model without them gives 14.9%. The five classical learners improve in the same way, and ridge regression remains the most accurate. For the ensemble and the tuned hybrid, whose out-of-fold errors by country were available as squared-error shares, the RMSE without their own six worst-fitted countries is 1.66 and about 1.9 t per person, against 1.57 for ridge regression without its own six. The low R^2 of the hybrids is therefore produced by a few countries, but excluding those countries does not change the ranking.

9. Line 103. *"ARMA/ARIMA" – Spell out abbreviations.*

Response. The sentence was removed when the Related work section was merged into the Introduction. All remaining abbreviations are written out at first use.

10. Lines 120–122. *"most non-Annex I countries rely on Tier-1 approaches with scarce activity data, and independent datasets disagree most where capacity is lowest" – To what extent do they disagree; how big is the error?*

Response. We now give figures with sources (Introduction, first paragraph): global totals of fossil CO_2 from different data sets differ by about 5% once system boundaries are harmonized, whereas estimates for individual countries, regions or fuels can differ by 15% or more (Andrew 2020); in Africa, two widely used data sets differ by less than 5% for most of the main emitters, but by more than 10% for South Africa and by a factor of three for Mali, and African countries do not report every year (Mostefaoui et al. 2024). On checking the sources we found that our original wording ("disagree most where capacity is lowest") was stronger than the evidence for fossil CO_2 supports, and we have replaced it with these statements.

11. Lines 126–127. *Many abbreviations; what do they all stand for?*

Response. The paragraph was removed in the merger of the Related work section. In the revised text every abbreviation is written out at first use (for example, particle swarm optimization, long short-term memory, gated recurrent unit, convolutional neural network), and the legends of Figs 1 and 4 and the footnotes of the tables repeat the definitions.

12. Line 135. *"The gap this study fills" – Maybe combine this with the paragraph from line 86?*

Response. Done. The two passages were combined and reduced to the four research questions and one sentence on what is new.

13. Lines 144–146. *Feels a bit repetitive.*

Response. The passage was removed.

14. Line 157. *"(constant international-$" – $ person^{-1}?*

Response. Yes. The unit is now given as "international dollars per person at 2011 prices" (Methods, *Data*; Table 1).

15. Line 160. *"filled by linear interpolation" – Is there another way to handle missing values, without having to use linear interpolation?*

Response. Yes, and examining this question changed our understanding of the missing values. The entries that were filled over long periods are not unknown: they are blanks for fuels that a country did not burn, as the accounting identity shows (see our response to Reviewer 1, Comment 9). The appropriate alternative is therefore to set them to zero, which we have done as a sensitivity analysis: the median error of ridge regression is 16.2% instead of 16.3% ($p = 0.19$). We also repeated the analysis without the countries with long filled runs and without the fuel shares altogether (Table 5). We considered multiple imputation for panel data (Honaker and King 2010) and did not use it, because it is designed for values that are unknown, which these are not; this is stated in the Methods (*Sensitivity analyses*).

16. Line 183. *"For Experiment C" – What about experiment A and B?*

Response. The three experiments are now introduced together, before they are used, in the new section *Study design and workflow* and in Fig 1.

17. Line 201. *Mean and standard deviation.*

Response. Added: "where μ_{tr} and σ_{tr} are the mean and the standard deviation of the covariate over the training countries of that fold" (after Eq 3).

18. Line 214. *"C = P·(G/P)·(E/G)·(C/E)" – Two different C's?*

Response. The notation was ambiguous, because *E* and *G* denoted per-capita quantities elsewhere and *c* denoted per-capita emissions. The identity is now written with separate symbols for national totals (*C* emissions, *P* population, *Y* GDP, *Q* primary energy; Eq 4), each defined in the text, and we note that *C* appears on both sides because the expression is an identity. Eq 5 then links the totals to the per-capita quantities used in the models.

19. Line 223. *Explain variables.*

Response. All symbols of the ridge equation (Eq 6) are now defined directly after it: the sample index, the number of training samples, the vector of the standardized window, the coefficients, the intercept and the penalty.

20. Line 237. *Please indicate what all parameters mean.*

Response. The equations of the LSTM, GRU, convolutional and attention layers are standard. To shorten the Methods we moved them to S1 Text (section C), where every symbol is now defined: the weight matrices, the bias vectors, the gates, the cell and hidden states and the activation functions.

21. Lines 299–300. *Also explain S, F, interpretation of phi_i.*

Response. The SHAP formula is now in S1 Text (section G) with a definition of every symbol (the set of all inputs, the subset S, its size, and the expected estimate given a subset) and an interpretation of the SHAP value (sign, sum over inputs, and mean absolute value as a measure of importance). The Methods (*Explainability*) describe the same in words.

22. Line 324. *"so country-years are far from independent" – Do you also account for this in selecting your training and test sets?*

Response. Yes. All years of a country are always in the same fold, and the inner validation sets used for early stopping, tuning and calibration are also formed by country. The Methods now say this explicitly in two places (*Estimation task and validation*, and *Model comparison and selection*: "The validation folds respect this structure, because all years of a country fall in the same fold, and the statistical comparison of two models must respect it as well").

23. Line 344 (Results). *There is a lot of methods, as well as interpretation in this section. This would fit better in the Methods and Discussion, respectively.*

Response. The Results were rewritten accordingly. Methods material (the description of the framework and the architecture, the imputation audit, the covariate ablation design, the rationale of the tests) is now in the Methods, and interpretive passages are in the Discussion.

24. Line 362. *"Figure 5. The Kaya-guided CNN-BiLSTM-Attention (KG-CBA) architecture" – Maybe this and previous figures in methods already?*

Response. Done. The workflow (Fig 1), the data (Fig 2) and the architecture (Fig 3) are now presented in the Methods. The workflow and architecture figures were redrawn.

25. Line 363. *"Experiment A:" – Use this terminology also in Methods.*

Response. Done; see comment 16. The terms Experiment A, B and C are defined in the Methods and used consistently afterwards.

26. Lines 437–439. *"The practical reading is direct: where energy-mix statistics exist, a single covariate is worth more than any architectural innovation evaluated here." – Seems like a statement for in the discussion.*

Response. The statement was moved to the Discussion (answer to Q3 in *Principal findings*, and *What limits accuracy: covariate data*).

27. Line 442. *"Table 3. Imputation audit" – Maybe put in the methods?*

Response. Done. The audit is now part of Table 1 in the Methods (*Data*), and the audit by country is in S1 Table.

28. Lines 462–463. *"so its apparently smooth covariate trajectory is substantially an artefact of the edge-extension rule rather than a measured series" – Is there a way to prevent this?*

Response. Yes: by setting the blank fuel shares to zero, which is what they are, or by omitting the fuel shares. On examination, the artefact we described is much smaller than we had thought. For Ghana, the filled values are a coal share of 0.001 after 1997 and a gas share of 0.002 before

2009, both practically zero, and the per-country median error of ridge regression is 13.0% with the original fill and 13.3% with zeros. The sentence was therefore removed, together with the related interpretation in the Discussion (see *Corrections made on our own initiative*, item 3).

29. Lines 479–480. *"This is an honest limitation of split-conformal inference under grouped sampling" – Maybe for in the discussion?*

Response. The interpretation was moved to the Discussion (Limitations; roadmap item 6). The Results now report only the coverage values, including the new analysis for ridge regression that shows the cause of the shortfall.

30. Line 483. *"SHAP attributions" – Did you try any other method of explainable AI? This can give different results.*

Response. We have now added two further measures that are defined directly on the error for held-out countries: permutation importance and drop-one-covariate importance, each for ridge regression and LightGBM (Methods, *Explainability*; Results, *Covariate importance*; Table 7). The three measures agree on the two most important covariates, energy per capita and the low-carbon share. They disagree in one instructive respect: for ridge regression, permutation importance assigns about 30 percentage points to each fuel share, although removing any one of them costs at most 1.1 points. The fuel shares are strongly related to one another, so that shuffling one of them produces combinations that do not occur. This confirms the reviewer's remark that methods can differ, and it is why we rely on the drop-one results for conclusions about the value of a covariate.

31. Lines 493–496. *Other than that SHAP and experiment C are defined differently, do you have an explanation for the difference in outcome in terms of what the data represents, the reality? What if you repeat experiment C for the most important parameter according to SHAP?*

Response. We followed the suggestion and repeated the experiment for the covariate that SHAP ranks first, energy per capita: without it, the median error of ridge regression rises from 7.7% to 29.4%, and that of LightGBM from 13.2% to 32.2% (Table 7). The loss is therefore largest for the covariate that SHAP ranks first and second-largest for the covariate that SHAP ranks second, the low-carbon share (+7.9 and +3.2 percentage points). There is thus no real difference in outcome, and our original wording, which suggested a contradiction, was misleading. In terms of what the data represent: energy per capita carries the scale of a country's emissions (the first term of the Kaya identity), and the low-carbon share carries the part of the carbon intensity of energy that the fossil-fuel shares cannot represent (the second term). Energy per capita is available for every country, so it does not limit the estimates; the low-carbon share is the most valuable covariate that many countries lack. This explanation is now given in the Discussion (*What limits accuracy: covariate data*).

32. Lines 497–498. *"The Experiment C contrast, not the SHAP ranking, is the evidence for the policy argument." – But how do you interpret the SHAP ranking then?*

Response. As a description of how much the fitted model relies on each input: energy per capita dominates because it sets the scale of emissions, the low-carbon share comes second, and the other covariates contribute little. This is consistent with the Kaya identity and with the results for held-out error (see comment 31). The sentence quoted was removed, and the interpretation is given in the Discussion.

33. Lines 550–551. *"Ensembling narrows this asymmetry but, on the evidence of clustered inference, does not beat ridge alone." – This message could be stated more clearly throughout the paper.*

Response. Agreed; see our response to Major comment 1. The message is now stated in the Abstract, Results, Discussion and Conclusions, and the ensemble is presented as exploratory.

34. Lines 553–555. *With this "observation level" you mean your Wilcoxon rank test?*

Response. Yes: the Wilcoxon signed-rank test applied to all 3,024 paired country-year errors, as opposed to the same test applied to the 112 per-country medians and to the cluster bootstrap. The three tests are now defined and named in the Methods (*Model comparison and selection*) and in the footnote of Table 4.

35. Line 562. *"sharpens the same point from an unexpected direction" – This I do not see; try to write it a bit more concrete.*

Response. The passage was removed, because the argument it introduced did not hold (see *Corrections made on our own initiative*, item 3). The Discussion now makes the point about data availability in concrete terms: the low-carbon share is available for 76 of the 112 countries of the panel and for only 4 of its 17 African countries, and the three African countries that are estimated worst are among those without it.

36. Line 613. *"Future work" – Since the linear model is not beaten by deep learning, you could maybe elaborate more on why this did not work, apart from data scarcity.*

Response. The new section *Why deep learning did not outperform the linear model* gives five reasons (see Major comment 1). Only one of them is the number of countries; the others are the near-linearity of the relationship, extrapolation to countries outside the training range, the fact that the remaining error is caused by missing information that no model can recover from the given covariates, and the variance added by tuning.

We thank the editor and the reviewers again for their time and for comments that have improved the manuscript considerably.

Yours sincerely,

Samiuel C. Atuahene, on behalf of all authors